

AI-Driven Demand Forecasting

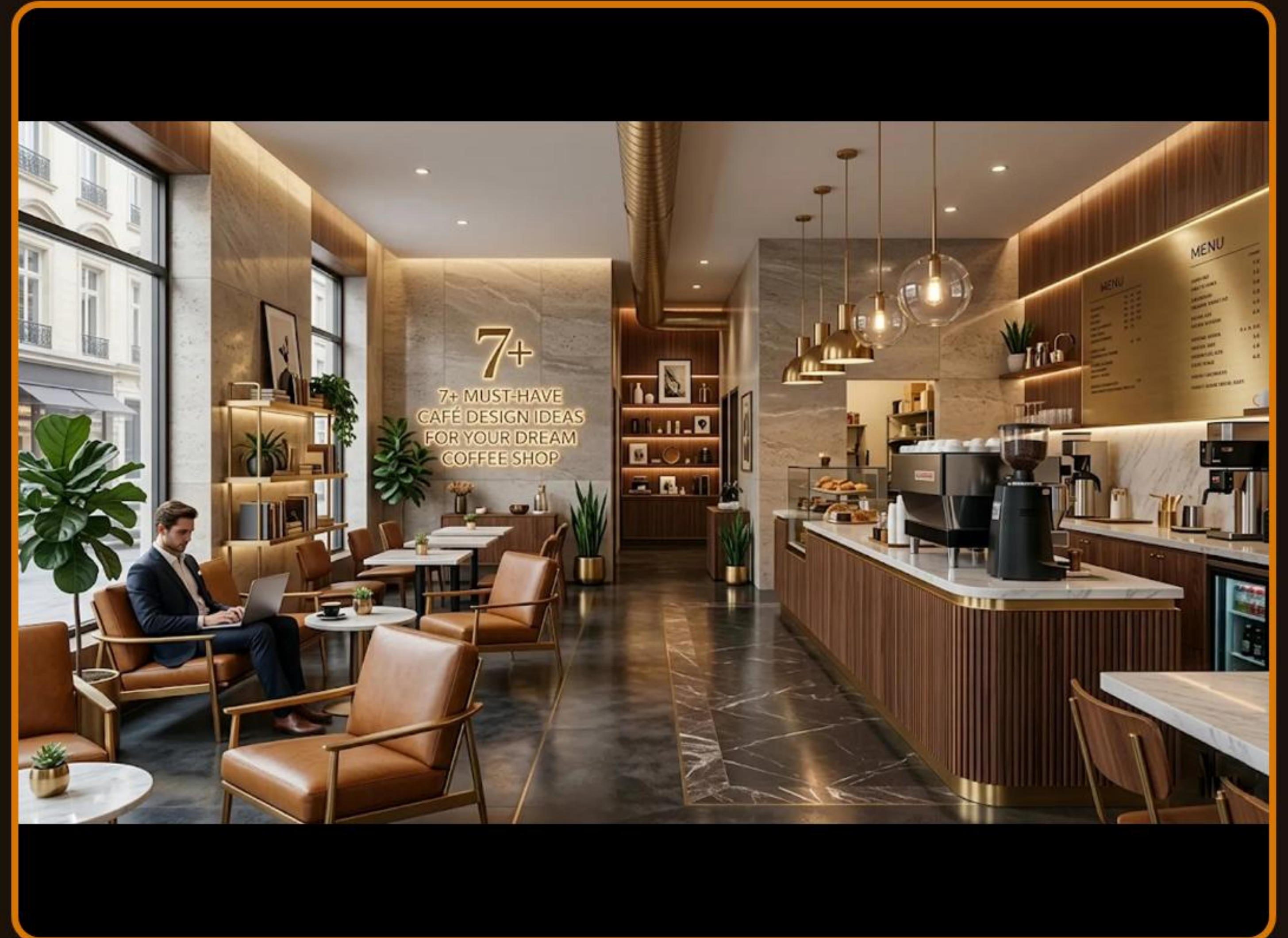
Optimizing Supply Chain Logistics

Kenangan Coffee • India

The Perishable Dilemma

High-volume O2O coffee chains face two critical extremes in inventory management:

- **Overstocking:** Leads to immense spoilage costs and environmental waste for milk and espresso beans.
- **Understocking:** Causes immediate revenue loss, stockouts, and high customer churn on the app.
- **The Barrier:** Traditional heuristics and manual ordering fail at retail scale.



Project Objectives



ETL Pipeline

Engineer an automated pipeline using Google BigQuery to aggregate daily POS and mobile app data.



Advanced ML

Train powerful models like XGBoost to predict 7-day forward demand at the hyper-local SKU level.



Waste Reduction

Decrease expired goods by >12% and reduce peak-hour stockout events by 20%.

Cloud System Architecture

Data Ingestion Layer

Real-time transactional data from the proprietary app and physical POS machines stream via Cloud Pub/Sub. BigQuery acts as our single source of truth for millions of historical rows.

MLOps & Serving

Vertex AI automates the training pipelines. Every midnight, Cloud Scheduler triggers a batch inference job, pushing the 7-day forecast straight to Looker Dashboards and SAP ERP.

XGBoost Implementation

The XGBoost Regressor effectively handles complex non-linear relationships in tabular time-series data.

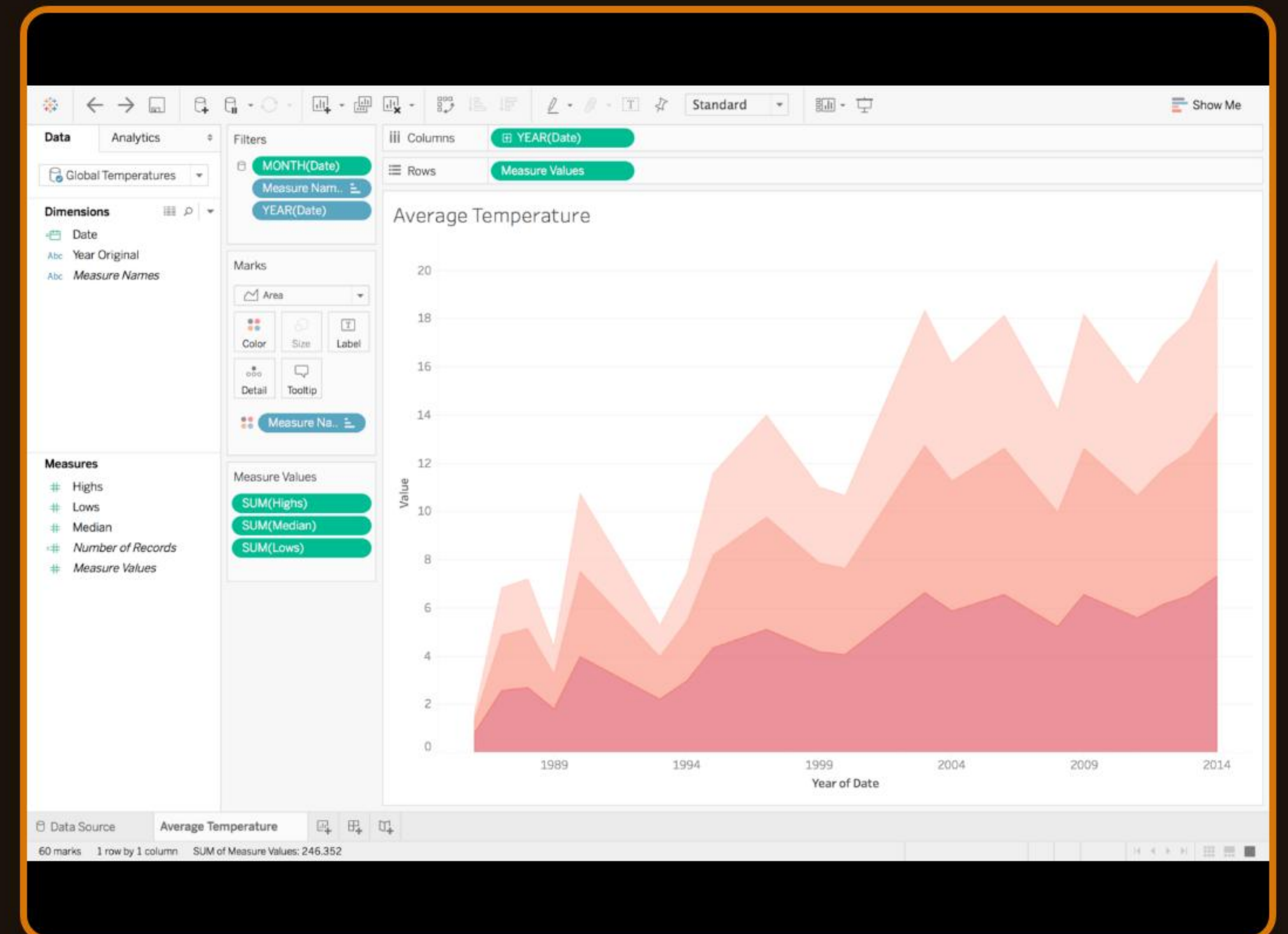
We explicitly engineered temporal lags and rolling window statistics to capture Indian retail momentum. TimeSeriesSplit ensures robust cross-validation without future data leakage.

```
48     best_idx = np.argmin(f_eps)
49     if f_eps[best_idx] < self._best_f:
50         self._best_f = f_eps[best_idx]
51         self._best_x = (self._x1 + self._x2) / 2
52
53     if self._is_mirror:
54         f_eps_mirror = np.apply_along_axis(
55             lambda x: self._f(x, self._best_x), 1,
56             self._X_mirror)
57         # check for best solution
58         best_idx = np.
59         if f_eps[best_idx] < self._best_f:
60             self._best_f = f_eps[best_idx]
61             self._best_x = (self._x1 + self._x2) / 2
62         f_eps = (f_eps + f_eps_mirror) / 2
63
64     f_eps = (f_eps - np.min(f_eps)) / (np.max(f_eps) - np.min(f_eps))
65     l_eps = np.mean(f_eps)
66     insert (md_eps.sha1, l_eps)
67     add docstring
```


Forecasting Results

The forecasting dashboard illustrates our model's pinpoint accuracy compared to actual demand.

- Baseline algorithms fail entirely during volatile Payday promotions.
- Our AI accurately maps non-linear spikes.
- Ensures stores are stocked precisely when demand surges, preventing costly stockouts.



Model Evaluation (MAPE)

7-Day Moving Avg

28.5%

Facebook Prophet

18.2%

XGBoost (Selected)

11.4%

MEAN ABSOLUTE PERCENTAGE ERROR EVALUATION

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right|$$

Business Impact

14%

WASTE REDUCTION

Massive ROI Achieved

By bringing the error rate down from 28.5% to 11.4%, the over-ordering of highly perishable items is drastically reduced.

We project a 14% reduction in total inventory waste value across NCR pilot stores, alongside a 20% drop in peak-hour stockouts.

Questions?

Thank you for your time and attention.

Satyam Kumar Singh | AI Engineering Intern

Image Sources



<https://static.gooecloud.com/upload/img/GY569396824781/7-must-have-cafe-design-ideas-to-elevate-your-coffee-shop.webp>

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